

AMOS OMAR NJAGI NDUNGO

Full-Stack Software Engineer | Data & ML Systems | Cybersecurity

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PROFESSIONAL SUMMARY

Full-stack Software Engineer with hands-on experience designing and shipping production-grade backend systems (FastAPI, PostgreSQL, Celery), machine-learning pipelines (scikit-learn, XGBoost), and modern web frontends (Next.js, React, TypeScript). Recent project work spans post-harvest supply-chain optimization, predictive maintenance for industrial equipment, and ML-driven risk scoring for social-impact programs. Complements engineering depth with formal training in malware analysis, penetration testing, and API security, plus a strong technical-writing and communication background built over several years of freelance content and media work. Looking to bring a build-and-break engineering mindset to a team building secure, data-driven, user-centric products.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Kotlin, Dart, SQL, x86 Assembly

Backend & APIs: FastAPI, Django, Flask, REST API design, SQLAlchemy (async), Pydantic, Celery, Redis

Frontend & Mobile: Next.js, React, Tailwind CSS, shadcn/ui, TanStack Query, Flutter, Android

Data & Machine Learning: Pandas, scikit-learn, XGBoost, SHAP, MLflow, NetworkX, Apache Kafka/Redpanda

Databases & Infrastructure: PostgreSQL, MySQL, Docker & Docker Compose, Nginx, HAProxy, Kubernetes & Cloud Native (Linux Foundation certified), Unix/Linux administration, Git/GitHub, Oracle Cloud

Cybersecurity: Web/mobile/API penetration testing, malware analysis & reverse engineering (APKs, binaries), vulnerability assessment, OSINT, security auditing & GRC, threat intelligence

Additional: Technical writing & documentation, video editing (DaVinci Resolve, Riverside.FM, Restream)

KEY PROJECTS

fresh-supplies — Post-Harvest Spoilage Prediction & Market Optimization Platform

Full-stack + ML platform | github.com/umarndungo/fresh-supplies

- Building a data-driven platform to reduce post-harvest losses across Sub-Saharan Africa through spoilage-risk prediction, market recommendation, and route optimization for perishable-crop supply chains.
- Architected the backend with FastAPI, async SQLAlchemy 2.0, and PostgreSQL 16 using a Clean Architecture pattern (domain, infrastructure, application, and API layers); built the frontend with Next.js 16, React 19, TypeScript, and Tailwind CSS.
- Developed a machine-learning data engine (scikit-learn, XGBoost, SHAP) for real-time spoilage-risk scoring, plus a mobile-first, offline-sync API with OTP authentication for field data collection.
- Implemented role-based access control for four user types (Administrator, Logistics Manager, Cooperative, Market Analyst) across 100+ commits, containerized with Docker Compose and targeting Oracle Cloud.

MaintainNexus — Predictive Maintenance & Work-Order Dispatch System

Backend, ML & mobile | github.com/umarndungo/maintainnexusai

- Built a predictive-maintenance and work-order dispatch system for industrial equipment (pumps, loading arms, valves), automating the pipeline from alert ingestion through technician dispatch and inventory checks.
- Engineered asynchronous alert processing with Celery and Redis, an XGBoost-based ML risk-scoring model, and append-only, hash-chained audit logging for compliance and traceability.
- Delivered live updates via Server-Sent Events and role-based APIs (Engineer, Supervisor, Executive) across a FastAPI backend, Next.js web dashboard, and Flutter mobile app, spanning 150+ commits.

Inuka Foundation Predictive Analytics — ML for Social Impact

72-hour hackathon, 5-person team | github.com/umarndungo/Inuka-Foundation-Predictive-Analytics

- Collaborated within a five-role team to build a predictive-intelligence system identifying program beneficiaries at risk of education dropout, within a 72-hour hackathon sprint.
- Designed a multi-tier (Bronze/Silver/Gold) data lake with Kafka/Redpanda event streaming and an explainable XGBoost risk-scoring model, triggering automated SMS outreach via Twilio and n8n for high-risk cases.
- Built an offline-first field data-entry system (IndexedDB queue) and a real-time dashboard (Next.js, MapLibre GL, Recharts) for program managers and field workers across five Kenyan regions.

PROFESSIONAL EXPERIENCE

EMTECHHouse — Software Development Apprentice (QA & Mobile)	<i>Jul 2026 – Present</i>
• Performed QA testing on an inventory and warehouse management system, identifying and documenting defects across functional and workflow test cases. • Contributed to mobile development on the inventory and warehouse management project and on a Learning Management System (LMS) mobile app. • Designed mobile UI/UX for a SACCO (savings and credit cooperative) management application. • Built mobile app companions for the fresh-supplies and MaintainNexus platforms, extending both projects' web and backend systems to mobile.	
M-Gas Kenya — Telephonic Interviewer Researcher	<i>Sep 2024 – Jan 2025</i>
• Conducted telephonic interviews to support market research initiatives and reported on findings across multiple projects. • Responded to customer inquiries on social media platforms and supported the social media team in analyzing reach and acquiring leads.	
Young and Ambitious in Africa Podcast — Host	<i>Aug 2022 – Apr 2024</i>
• Hosted a podcast interviewing African thought leaders about their career journeys across a range of fields. • Owned end-to-end production: editing, scheduling, and content management.	
Boost Technology — Independent Content Writer	<i>Jul 2022 – Feb 2023</i>
• Helped launch the “Founder Series” on the company blog and contributed articles, video edits, and technical documentation.	
Freelance Writing — Ghost Writer	<i>Dec 2020 – Jul 2022</i>
• Wrote and edited English-language blog content, articles, and technical documentation for industrial companies across Asia; also handled video editing.	
KenGen — ICT Industrial Attachment	<i>May 2019 – Jul 2019</i>
• Supported the ICT department with software installation, updates, and deployment (operating systems, antivirus), plus hardware maintenance and general system administration.	

CERTIFICATIONS & TRAINING

• Cybersecurity Swara Course — Africa Hackon Academy Certified Mar 2026: penetration testing & defense (web/mobile app security, network security, OSINT), vulnerability assessment & mitigation, GRC & cloud security. • Malware Analysis & Introduction to Assembly Language — IBM / Coursera Certified Mar 2026: malware analysis, reverse engineering, and low-level systems for threat analysis. • API Penetration Testing — APISec University In progress: advanced API security assessment and secure implementation strategies. • Android Hacking Map — Hextree & Google In progress: Android security research, reverse engineering, and mobile vulnerability analysis. • Kubernetes and Cloud Native Essentials — The Linux Foundation 2025	
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EDUCATION

BSc. Computer Science and Technology — Meru University of Science and Technology	<i>2016 – 2021</i>
Software Engineering — ALX Africa (Holberton School)	<i>2021 – 2022</i>
Python Development Accelerated Learning Camp — Andela 21	<i>Feb 2016</i>